

# CHE 201 Spring 2020 Midterm 1

Anna Christensen

TOTAL POINTS

**93 / 100**

## QUESTION 1

### Question 1 20 pts

#### 1.1 a 10 / 10

- ✓ - **0 pts** Correct
- **2 pts** first is incorrect
- **2 pts** second is incorrect
- **2 pts** third is incorrect
- **2 pts** fourth is incorrect
- **2 pts** fifth is incorrect

#### 1.2 b 4 / 5

- **0 pts** Correct
- **2 pts** gauge pressure not converted to absolute
- **2 pts** T not converted to K
- ✓ - **1 pts** P1 is wrong

#### 1.3 C 5 / 5

- ✓ - **0 pts** Correct
- **5 pts** incorrect

## QUESTION 2

### Question 2 50 pts

#### 2.1 a 3 / 3

- ✓ - **0 pts** Correct

#### 2.2 b 8 / 8

- ✓ - **0 pts** Correct
- **4 pts** Pv incorrect
- **4 pts** Tv incorrect
- **7 pts** States not labeled in each
- **2 pts** pv state 1 and state 3 incorrect
- **1 pts** tv state 3 incorrect
- **2 pts** both graphs state 1 is wrong
- **2 pts** both graphs state 2 is wrong

#### 2.3 C 5 / 5

- ✓ - **0 pts** Correct

#### 2.4 d 2 / 2

- ✓ - **0 pts** Correct

#### 2.5 e 11 / 15

- **0 pts** Correct
- **3 pts** v1 is incorrect or not calculated
- **3 pts** v2 is incorrect
- ✓ - **2 pts** not multiplied with mass
- ✓ - **2 pts** units are not converted
- **15 pts** not done

#### 2.6 f 15 / 15

- ✓ - **0 pts** Correct
- **3 pts** u1 is incorrect or not calculated
- **3 pts** u3 is incorrect or not calculated
- **15 pts** incorrect
- **2 pts** x is wrong
- **2 pts** u not multiplied with mass
- **2 pts** calculation error
- calculation error because of work

#### 2.7 g 2 / 2

- ✓ - **0 pts** Correct
- **2 pts** incorrect or not done
- **0.5 pts** s1 is wrong
- **1 pts** s3 is wrong
- **0.5 pts** s2 is wrong

## QUESTION 3

### Question 3 30 pts

#### 3.1 a 2 / 2

✓ - 0 pts Correct

### 3.2 b 2 / 2

✓ - 0 pts Correct

- 0.5 pts ideal gas is missing
- 0.5 pts closed system is missing

### 3.3 C 1 / 3

- 0 pts Correct

✓ - 2 pts Z ideal and Znotideal are not correct

- 1 pts Z for not ideal gas is not correct
- 3 pts wrong

### 3.4 d 3 / 3

✓ - 0 pts Correct

- 3 pts incorrect
- 1 pts line should be not linear
- 1 pts P and V are not marked

### 3.5 e 10 / 10

✓ - 0 pts Correct

- 1 pts calculation error
- 10 pts incorrect or not done

### 3.6 f 10 / 10

✓ - 0 pts Correct

- 10 pts incorrect or not done
- 1 pts calculation error

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Please fill in answers in the space supplied and show all your work to receive full credit.  
Useful Tables at end of Exam.

1. a) (10 pts) State whether the following statements are **True or False**

- F An ideal gas at a given state expands to a fixed final volume at constant pressure or at constant temperature. Work is smaller in constant pressure case.
- T A two-phase liquid-vapor mixture with equal volumes of saturated vapor and saturated liquid has a quality of less than 0.5.
- T As temperature decreases from the critical temperature, the values of the specific volume of saturated vapor and saturated liquid moves apart from each other.
- F Internal energy is an intensive property.
- F Specific volume of compressed liquid can be approximated to specific volume of saturated liquid at the same pressure.

b) (5 pts) The gage pressure of an automobile tire is measured to be 30psig before a trip and 32psig after the trip at a location where the atmospheric pressure is 14psia. Assuming the volume of the tire remains constant and the air temperature before the trip is 25°C, determine the air temperature in the tire after the trip. (Air is ideal gas)

$$\begin{aligned}
 P_{\text{gage}} &= P_{\text{absolute}} - P_{\text{atm}} & PV &= nRT & P_{\text{gage1}} &= 30 \text{ psig} \\
 P_2 &= 32 + 14 = 46 \text{ psia} & \frac{P_1}{T_1} &= \frac{P_2}{T_2} & P_{\text{gage2}} &= 32 \text{ psig} \\
 P_1 &= 30 + 14 = 31 \text{ psia} & T_2 &= \frac{P_2 T_1}{P_1} = \frac{(46)(298)}{31} = 442 \text{ K} & P_{\text{atm}} &= 14 \text{ psia} \\
 & & & & T_1 &= 25^\circ\text{C} \rightarrow 298 \text{ K} & T_2 &=?
 \end{aligned}$$

c) (5pts) What would be the specific volume of saturated solid carbon dioxide at the same temperature, greater or less than saturated vapor value? Why?

The specific volume of saturated solid would be less than the specific volume of saturated vapor because the density of the solid is greater.  $\text{specific volume} = \frac{1}{\text{density}}$   
so as density increases, specific volume decreases.

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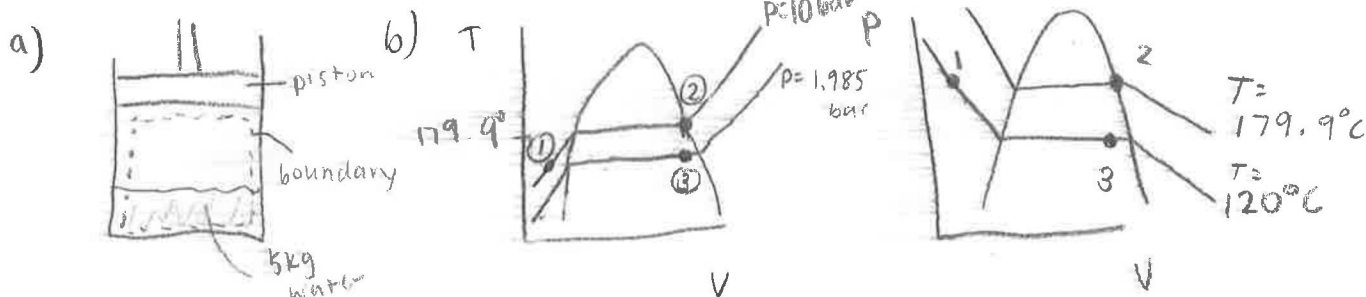
2. (50 pts) 5kg of water at contained in a piston-cylinder assembly undergoes two processes in series from an initial state where the temperature is 120°C and 10 bar.

$T_1 = 120$   $P_1 = 10 \text{ bar}$

- **Process 1–2** System is heated at constant pressure to a saturated vapor state.
- **Process 2–3** System is cooled at constant volume until it reaches to 120°C.

Kinetic and potential effects can be ignored.

- (3 pts) Draw a schematic of the system and label the system boundary.
- (8 pts) Carefully sketch the process on a:
  - P-v diagram (Pressure-specific volume)
  - T-v diagram (Temperature-specific volume)
- (5 pts) State the engineering model
- (2 pts) Write the simplified energy balance
- (15 pts) Calculate the work transferred in kJ. Is work done on the system or by the system?
- (15 pts) Calculate the heat transferred in kJ. Is heat transferred to the system or from the system?
- (2 pts) Calculate the mass of vapor in each state in kg.



c) model

- closed system
- work is in  $\Delta V$  form
- $KE = PE = 0$

d)  $\Delta U = Q - W$

e) 

| 1                           | 2            | 3                       |
|-----------------------------|--------------|-------------------------|
| $T = 120$                   | $T = 179.9$  | $T = 120$               |
| $P = 10$                    | $P = 10$     | $P = 1.985 \text{ bar}$ |
| $v = 1.0603 \times 10^{-3}$ | $v = 0.1944$ | $v = 0.1944$            |
| $u = 503.50$                |              |                         |

$W_{12} = \int P dv$   $P = 10 \text{ bar}$

$W = 10 \text{ bar} (0.1944 - 1.0603 \times 10^{-3} \frac{\text{m}^3}{\text{kg}})$

$W_{12} = 1.933 \text{ kJ}$

$W_{23} = 0$

$W = 1.933 \text{ kJ}$

work is done by the system

at  $T = 120$

$u_f = 503.50$   $u_g = 2529.3$

$v = (1-x)v_f + xv_g$   
 $0.1944 = (1-x)(1.0603 \times 10^{-3}) + x(0.8919)$

$0.1933 = 0.8908x \rightarrow x = 0.2170$

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f)

$$u = (1-x)u_f + x u_g$$

$$u = (1-0.2170)(503.5) + 0.217(2529.3)$$

$$u_3 = 943.1 \frac{\text{kJ}}{\text{kg}}$$

$$\Delta u = Q - W$$

$$m(u_3 - u_1) + W = Q$$

$$u_1 = 503.5$$

$$Q = (5 \text{ kg}) (943.1 - 503.5 \frac{\text{kJ}}{\text{kg}}) + 1.933 \text{ kJ} = 2.1999 \times 10^3 \text{ kJ}$$

heat is transferred from the system

g)

state 1 = 0 kg → just liquid 0% vapor

state 2 = 5 kg → just vapor 100%

state 3 = 1.085 kg

$$x = \frac{m_{\text{vapor}}}{m_{\text{total}}} = \frac{m_{\text{vapor}}}{5 \text{ kg}} = 0.217$$

$$m_{\text{vapor}} = 1.085$$

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3. (30 points) A piston cylinder device initially contains  $0.5\text{m}^3$  of air at  $80\text{kPa}$  and  $200^\circ\text{C}$ . The air is now compressed to  $0.1\text{m}^3$  in such a way that the temperature inside the cylinder remains constant. The air is modeled as an ideal gas and kinetic and potential energy effects are negligible.
- (2 pts) Draw a schematic of the system and label the system boundary.
  - (2 pts) State the engineering model
  - (3 pts) Write the simplified energy balance and calculate compressibility factor ( $Z$ ) for ideal gas and not ideal gas state 1.
  - (3pts) Draw P-v diagram (Pressure-specific volume)
  - (10 pts) Determine the work (kJ)
  - (10 pts) Determine the heat (kJ)

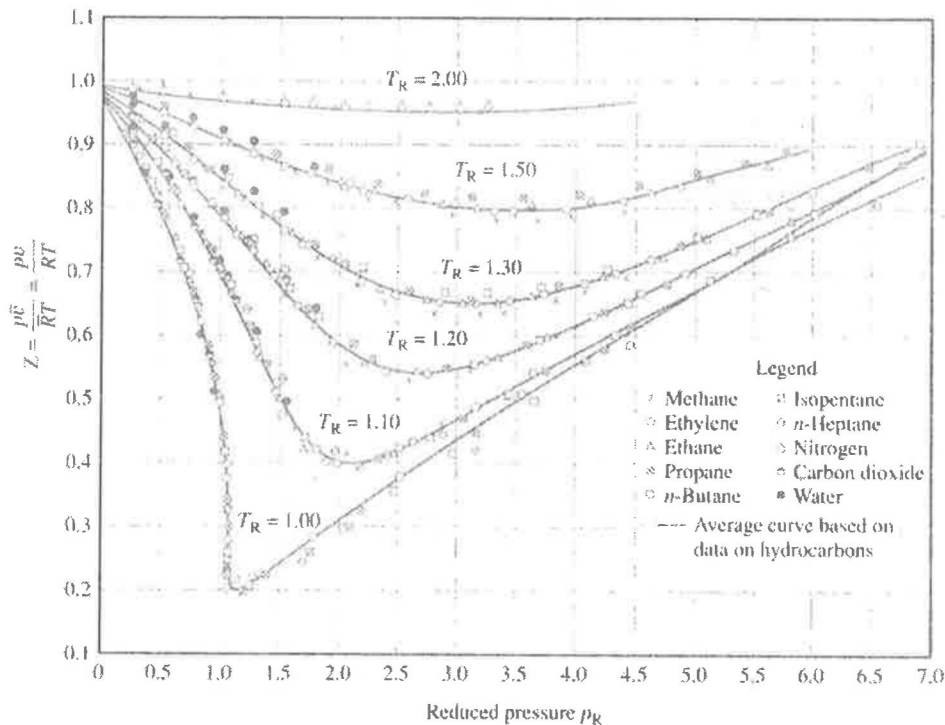
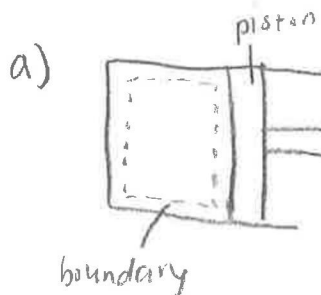


FIG. 3.12 Generalized compressibility chart for various gases.



- b) model
- closed system
  - $KE = PE = 0$
  - Temperature is constant
  - Work is in form of  $\Delta V$
  - ideal gas

$$V_1 = 0.5\text{m}^3$$

$$P_1 = 80\text{kPa}$$

$$T_1 = 200^\circ\text{C} = T_2$$

$$V_2 = 0.1\text{m}^3$$

$$P_2 = ?$$

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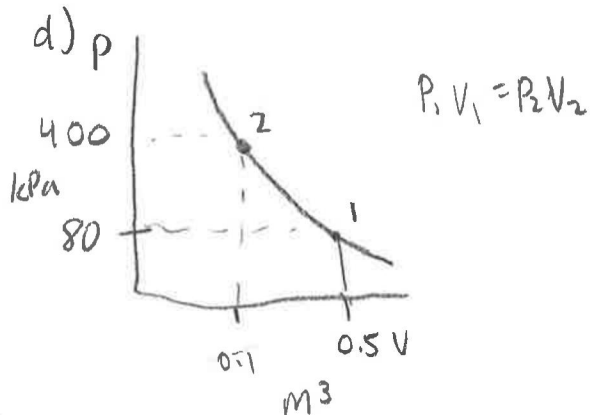
c)  $\Delta U = Q - W$

at  $T = 200^\circ$

$P_r = 0.3363$

$V_r = 1707$

$$Z = \frac{P_c V_c}{R T_c} = 0.284 = Z$$



e)

$$W = \int P dv \quad P_1 V_1 = RT$$

$$= \int_{V_1}^{V_2} \frac{RT}{V} dv = RT (\ln V_2 - \ln V_1) = RT \ln \left( \frac{V_2}{V_1} \right) = P_1 V_1 \ln \left( \frac{V_2}{V_1} \right)$$

$$W = (80 \text{ kPa})(0.5 \text{ m}^3) \ln \left( \frac{0.1}{0.5} \right) = \boxed{-64.38 \text{ kJ}}$$

f)  $\Delta U = Q - W$

$$Q = \Delta U + W = m(u_2 - u_1) - W$$

$u_2 = 142.56$

$\Delta U = 0$

$u_1 = 142.56$

so

$Q = W$

$$\boxed{Q = -64.38 \text{ kJ}}$$

